

Mockito Hands-On Exercises

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Exercise 1: Mocking and Stubbing

Problem Analysis: Test a service that depends on an external API without calling the real API.

Root Cause: Direct dependency on external services makes testing slow and unreliable.

Solution:

1. Create a mock object using Mockito.
2. Stub `getData()` to return 'Mock Data'.
3. Inject the mock into `MyService` and verify the returned value using `assertEquals()`.

Expected Output:

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS

Prevention Tips:

- Mock external dependencies.
- Keep unit tests independent.
- Stub only required methods.

Exercise 2: Verifying Interactions

Problem Analysis: Ensure that dependent methods are invoked correctly.

Root Cause: Without interaction verification, required method calls may be missed.

Solution:

1. Create a mock object.
2. Call `fetchData()`.
3. Verify `getData()` is invoked using `verify(mockApi).getData()`.

Expected Output:

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS

Prevention Tips:

- Verify only important interactions.
- Avoid unnecessary verifications.
- Keep tests focused on one behavior.

Conclusion

Mockito simplifies unit testing by creating mock objects, stubbing method behavior, and verifying interactions. It helps isolate business logic from external dependencies, making tests reliable, maintainable, and fast.